

1. Title

EcoTrack: AI-Powered Carbon Footprint & Sustainability Management Platform

A full-stack sustainability management platform that helps individuals and organizations monitor, analyze, and reduce their carbon footprint through activity tracking, AI-driven recommendations, sustainability goals, challenges, and environmental analytics.

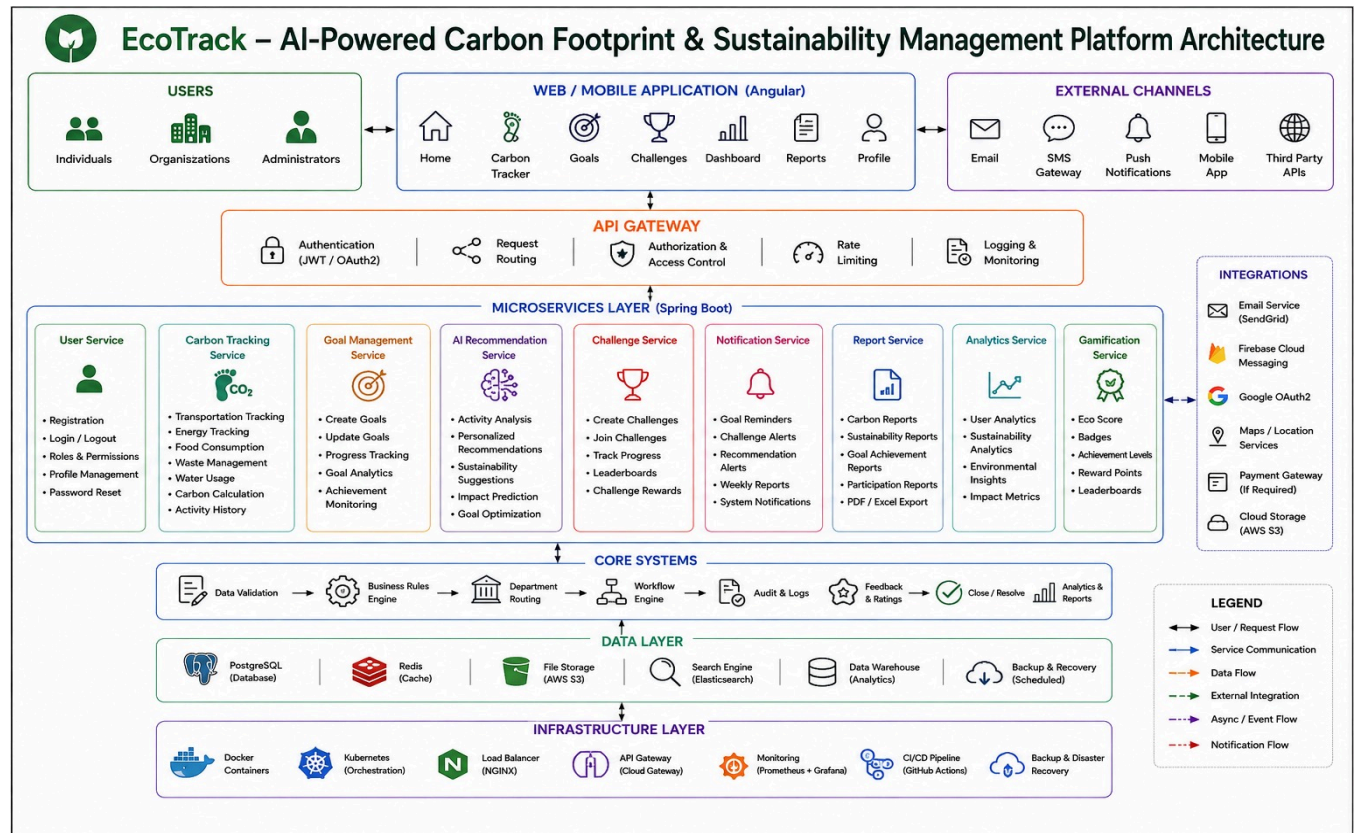
2. Objective

Build a full-stack web application (Angular frontend + Spring Boot backend) that enables users to calculate carbon emissions from daily activities, track sustainability goals, participate in eco-friendly challenges, receive personalized recommendations, and generate environmental impact reports.

Outcomes

- (i) Developed and deployed a full-stack Angular + Spring Boot application.
- (ii) Implemented secure user authentication using JWT and OAuth2.
- (iii) Built carbon footprint calculation and tracking functionality.
- (iv) Implemented sustainability goal management.
- (v) Developed AI-powered environmental recommendation system.
- (vi) Added community challenges and gamification features.
- (vii) Created dashboards for users and administrators.
- (viii) Implemented notification and reporting systems.
- (ix) Deployed frontend and backend using PostgreSQL in production.
- (x) Added sustainability analytics and environmental monitoring systems.

3. Architecture Diagram



4. Modules to be Implemented

1. User Authentication & Role-Based Access

- JWT Authentication
- OAuth2 Login
- Password Reset
- Profile Management

Roles

User

Administrator

2. User Profile Management

- (i) Profile Creation
- (ii) Profile Picture Upload
- (iii) Sustainability Preferences
- (iv) Location Management
- (v) Environmental Interests
- (vi) Personal Goal Preferences
- (vii) Lifestyle Configuration

Interest Categories

Renewable Energy

Recycling

Waste Reduction

Sustainable Living

Green Transportation

Water Conservation

Eco-Friendly Products

Climate Action

Organic Farming

Wildlife Conservation

3. Carbon Footprint Tracking System

- (i) Transportation Tracking

- (ii) Energy Consumption Tracking
- (iii) Food Consumption Tracking
- (iv) Waste Management Tracking
- (v) Water Usage Tracking
- (vi) Carbon Emission Calculation
- (vii) Historical Activity Records

Tracking Categories

Transportation

Electricity Usage

Fuel Consumption

Food Consumption

Waste Generation

Water Usage

Online Shopping

Travel Activities

4. Sustainability Goal Management

- (i) Create Goals
- (ii) Update Goals
- (iii) Goal Progress Tracking
- (iv) Goal Achievement Monitoring
- (v) Goal Analytics

Goal Types

Reduce Carbon Emissions

Reduce Electricity Consumption

Reduce Water Usage

Increase Recycling

Reduce Plastic Usage

Use Public Transport

Plant Trees

5. AI Recommendation System

(i) User Activity Analysis

(ii) Personalized Recommendations

(iii) Sustainability Improvement Suggestions

(iv) Environmental Impact Prediction

(v) Goal Optimization

Recommendation Factors

Carbon Emissions

Lifestyle Habits

Energy Usage

Transportation Patterns

Waste Management

Goal History

6. Community Challenge Management

(i) Create Challenges

(ii) Join Challenges

(iii) Track Challenge Progress

(iv) Leaderboards

(v) Challenge Rewards

Challenge Categories

Plastic-Free Week

Cycle To Work

Energy Saving Challenge

Tree Plantation Drive

Water Conservation Week

Zero Waste Challenge

7. Search & Discovery System

(i) Search Challenges

(ii) Search Users

(iii) Filter By Sustainability Interests

(iv) Filter By Location

(v) Filter By Environmental Goals

(vi) Recommended Challenges

8. Notification System

(i) Goal Reminders

(ii) Challenge Notifications

(iii) Recommendation Alerts

(iv) Weekly Sustainability Reports

(v) System Notifications

9. Dashboard & Analytics

User Dashboard

(i) Carbon Footprint Overview

(ii) Sustainability Score

(iii) Goal Progress

(iv) Challenge Participation

(v) Environmental Impact Metrics

Admin Dashboard

(i) User Management

(ii) Challenge Moderation

(iii) Sustainability Analytics

(iv) Reports Management

(v) System Monitoring

10. Reports & Export System

(i) Carbon Footprint Reports

(ii) Goal Achievement Reports

(iii) Sustainability Reports

(iv) Challenge Participation Reports

(v) PDF Export

(vi) Excel Export

11. Gamification & Rewards System

- (i) Reward Points
- (ii) Sustainability Badges
- (iii) Achievement Levels
- (iv) Eco Score System
- (v) Leaderboards

Badge Categories

Green Beginner

Eco Warrior

Sustainability Champion

Climate Hero

Planet Protector

12. Final Integration, Testing & Deployment

5. Milestone 1: Week 1 & 2 — Requirements, Database Design & Backend Setup

Tasks

(i) Define project scope and user roles.

(ii) Design database schema:

users

profiles

carbon_entries

goals

goal_progress

challenges

notifications

reports

(iii) Initialize Spring Boot project.

(iv) Configure PostgreSQL/MySQL.

(v) Implement JWT authentication.

(vi) Create Angular frontend skeleton.

Outcomes

(i) Backend and frontend architecture setup completed.

(ii) Authentication flow functional.

(iii) Database schema finalized.

(iv) Role-based access configured.

6. Milestone 2: Week 3 & 4 — Carbon Tracking & Goal Management

Tasks

(i) Implement carbon tracking APIs.

(ii) Build carbon calculation engine.

(iii) Develop goal management functionality.

(iv) Create sustainability dashboard.

(v) Implement search and filter systems.

Outcomes

- (i) Carbon tracking operational.
- (ii) Goal management functional.
- (iii) Sustainability calculations completed.

7. Milestone 3: Week 5 & 6 — AI Recommendations & Community Challenges

Tasks

- (i) Implement recommendation engine.
- (ii) Build challenge management system.
- (iii) Develop gamification features.
- (iv) Create notification services.
- (v) Add leaderboard functionality.

Outcomes

- (i) Recommendation system operational.
- (ii) Community challenges functional.
- (iii) User engagement modules completed.

8. Milestone 4: Week 7 & 8 — Analytics, Testing & Deployment

Tasks

- (i) Build analytics dashboard.

- (ii) Add charts and reports.
- (iii) Implement testing and validations.
- (iv) Deploy frontend/backend using AWS, Render, or Vercel.
- (v) Configure SSL and production environment.

Outcomes

- (i) Fully deployed production-ready application.
- (ii) Analytics and reporting systems operational.
- (iii) Complete sustainability management workflow demonstrable.

9. Evaluation Criteria

Milestone 1 (Week 2)

- (i) Spring Boot project setup completed.
- (ii) JWT authentication implemented.
- (iii) Database schema finalized.
- (iv) Frontend authentication flow working.

Milestone 2 (Week 4)

- (i) Carbon tracking operational.
- (ii) Goal management functional.
- (iii) Sustainability dashboard implemented.

Milestone 3 (Week 6)

- (i) Challenge management operational.
- (ii) Recommendation system functional.
- (iii) User engagement workflow completed.

Milestone 4 (Week 8)

- (i) Fully deployed frontend and backend.
- (ii) Dashboard and reporting systems operational.
- (iii) End-to-end sustainability management workflow demonstrated.

10. Tools & Tech Stack

Programming Language

Backend

- (i) Java

Frontend

- (i) TypeScript
- (ii) JavaScript

Frameworks

Backend

- (i) Spring Boot
- (ii) Spring Security
- (iii) Spring Data JPA
- (iv) Hibernate
- (v) Maven

Frontend

- (i) Angular
- (ii) Angular Material
- (iii) RxJS

(iv) Chart.js

(v) Bootstrap

Database

Local Development

(i) MySQL

Production

(i) PostgreSQL

Authentication

(i) Spring Security

(ii) JWT Authentication

(iii) OAuth2 (Google Login)

Notifications

(i) Firebase Cloud Messaging

(ii) JavaMailSender

Storage

(i) AWS S3

(ii) Cloudinary

AI Services

(i) OpenAI API

(ii) Spring AI

Testing

(i) JUnit

(ii) Mockito

(iii) Postman

(iv) Jasmine

(v) Karma

Dev & Deployment Tools

(i) IntelliJ IDEA / VS Code

(ii) Git & GitHub

(iii) Docker

(iv) Docker Compose

(v) GitHub Actions

(vi) AWS / Render / Railway

(vii) Postman